

Supplement: Lineage Tables for the 10,240-Byte Generator

The DOPE Project

Abstract—Lineage-level retention and byte charges for the displayed profile, the 8,192-step sufficiency budget, and the 142 BeyondArena family names. These tables are the rows behind the medians in the main paper. A blank retention is an unavailable fit or an auditor that is not informative on all three sample seeds. It is not a zero. BeyondArena is not pooled with the PMLB panel. Master Fitness v2 is not in these tables.

APPENDIX A

DISPLAYED PROFILE, ALL 100 LINEAGES

TABLE I: Displayed profile, all 100 lineages.

id	name	rows	feat.	bytes	L3	CB n	CB $4n$	lin $4n$	MLP $4n$
02a45777900441ce	215_2dplanes	800	10	2,583	pass	0.9930	1.0001	0.9990	1.0477
047e62b52a3f4ede	695_chatfield_4	156	12	3,311	pass	1.0041	1.0155	1.0048	—
04c80944e971cb3e	first_principles_supernovae_zr	157	1	762	pass	0.8753	0.8846	0.9959	1.2627
0561d673cb73c27e	666_rmftsa_ladata	339	10	2,921	pass	1.0095	1.1026	1.0338	1.1394
060b9fff8092db17	589_fri_c2_1000_25	667	25	5,717	pass	0.8893	0.9534	0.9911	1.5794
06b0312a585b436e	1196_BNG_pharynx	800	10	2,631	pass	0.8601	0.8964	1.0458	1.7049
0779d3fd5b3133a2	nikuradse_2	241	1	758	pass	0.9096	0.9261	2.1723	2.2082
07f38946316c039b	593_fri_c1_1000_10	667	10	2,876	pass	0.5650	0.6068	0.9617	1.6005
0a8428ef25a69175	537_houses	800	8	2,609	pass	0.6257	0.7036	1.0150	1.0256
0d482c1dd7cc6592	210_cloud	72	5	1,844	pass	1.0374	0.9772	1.0944	1.0239
0db59fd2579600d0	665_sleuth_case2002	98	6	1,962	pass	1.0371	1.1324	1.1345	—
0f86a714ec5f3ac6	1191_BNG_pbc	800	18	4,183	pass	0.8097	0.8761	0.9173	—
167578f5deba4977	547_no2	333	7	2,443	pass	0.8073	0.8513	1.0097	0.9898
19f2c3c2d67defd7	feynman_II_11_27	800	4	1,494	pass	0.9704	0.9890	0.9960	1.0048
19f4780b53b3fa41	678_visualizing_environmental	74	3	1,515	pass	—	—	—	0.4431
1a4456d1cc71a26a	feynman_test_1	800	7	2,191	pass	1.0608	1.1671	0.9816	—
1a803b27c2400dc4	feynman_III_10_19	800	4	1,520	pass	0.9882	0.9952	0.9942	0.9965
1cb97b1d7314d43d	229_pwLinear	133	10	2,587	pass	0.8417	0.9878	0.9104	—
205f4b4982b9ea49	608_fri_c3_1000_10	667	10	2,845	pass	0.5263	0.5671	1.0451	0.8182
26eff843be35d831	581_fri_c3_500_25	333	25	5,647	pass	0.7602	0.9020	1.1297	1.5792
282f8ddb43b5e677	595_fri_c0_1000_10	667	10	2,904	pass	0.8974	0.9234	0.9962	1.0250
297b7688a045c583	603_fri_c0_250_50	167	50	10,171	pass	0.8848	1.1562	1.0700	2.6883
2c8d0985f3557d2b	feynman_II_11_28	800	2	1,026	pass	0.9831	0.9899	1.0031	0.9976
30ca2887550f0d8d	201_pol	800	26	8,145	pass	0.5742	0.5976	0.7015	0.6923
3408a6a513c627a3	strogatz_barmag1	267	2	1,011	pass	0.9750	0.9775	0.9969	1.0058
35a9aa9b88f0e4d1	344_mv	800	10	2,737	pass	0.9985	1.0016	0.9662	1.0347
3d3321e27d14f9a4	599_fri_c2_1000_5	667	5	1,707	pass	0.8359	0.8611	0.9668	1.6797
44eaabe4cb923766	613_fri_c3_250_5	167	5	1,712	pass	0.8671	0.9276	0.8599	—
46b25db9396c8209	227_cpu_small	800	12	3,327	pass	0.8569	0.9072	0.9018	0.9364
46c5a59e5ae42037	1193_BNG_lowbw	800	9	2,436	pass	0.9635	0.9703	1.0030	0.9871
471f11cc2d38853a	feynman_test_11	800	4	1,481	pass	0.9645	0.9760	0.9975	0.9826
47c581af5de1136b	582_fri_c1_500_25	333	25	5,670	pass	0.7510	0.9088	0.7173	—
48b7a3bf81e5f3e5	605_fri_c2_250_25	167	25	5,692	pass	0.4046	0.6465	1.9138	—
4a354d98c965b01f	597_fri_c2_500_5	333	5	1,713	pass	0.7550	0.8055	0.9814	1.8381
4aa84ff8ad0cd26	feynman_III_13_18	800	4	1,501	pass	0.8748	0.9137	0.8850	0.9348
51730b348b1bbd14	557_analcatdata_apnea1	316	3	1,237	pass	-0.0032	0.0086	—	—
54b221668673164c	feynman_III_14_14	800	5	1,726	pass	1.0346	1.1164	1.0011	1.0611
552665868f8f72ed	strogatz_vdp2	267	2	989	pass	0.9945	0.9963	0.9960	0.9969
585fef2a1cb6b33f	strogatz_lv2	267	2	1,018	pass	0.8286	0.9859	0.7502	—
642738a361f1c63a	feynman_I_12_1	800	2	1,032	pass	0.9892	0.9910	0.9892	0.9933
64c64f0f9983c48f	strogatz_bacres2	267	2	998	pass	0.9793	0.9832	0.9783	0.9862
654566a1f91408aa	4544_GeographicalOriginalofMusic	706	117	22,233	fail	—	—	—	—
693890aae9dc860b	first_principles_supernovae_zg	162	1	761	pass	0.8492	0.8908	1.0066	1.1997
6cacb11611345a24	feynman_I_11_19	800	6	1,947	pass	0.9524	0.9812	0.9780	0.9800
7291b347fe0238fb	feynman_II_11_20	800	5	1,757	pass	0.9928	1.0432	1.0016	1.0674
7423783f7e38bb9b	strogatz_shearflow1	267	2	1,021	pass	0.4900	0.5614	-0.3978	—
7a3d123dd555110d	598_fri_c0_1000_25	667	25	5,668	pass	0.8613	0.9166	1.0046	1.0034
7c8943b160464add	601_fri_c1_250_5	167	5	1,719	pass	0.7859	0.8552	1.0773	—
7e51711baa1b1fb1	feynman_I_12_11	800	5	1,718	pass	0.9267	0.9320	1.0015	0.9860
80589368fa78f21a	218_house_8L	800	8	2,392	pass	0.5835	0.6708	0.9616	1.1040
8164950b155f7f56	feynman_I_12_2	800	4	1,511	pass	0.9702	0.9986	0.9485	0.9753
85a31ae409944a1b	feynman_II_10_9	800	3	1,269	pass	0.9330	0.9508	0.9589	0.9515
8d859cfa7b51a351	feynman_III_15_12	800	3	1,248	pass	0.5172	0.5542	1.0228	1.0749
923483635406b213	1203_BNG_pwLinear	800	10	2,581	pass	1.2079	1.1750	0.9941	1.5230
9312d813229df035	690_visualizing_galaxy	216	4	1,458	pass	0.9725	0.9755	0.9853	1.0870
94a00ea67f49bf84	596_fri_c2_250_5	167	5	1,698	pass	0.7546	0.8072	—	—
94bd4520b8ceb3f0	feynman_test_12	800	5	1,747	pass	0.9688	0.9804	0.9913	1.0067
956d34f13e7af830	586_fri_c3_1000_25	667	25	5,701	pass	0.9325	0.9849	1.2339	2.1457
9a10c15885022034	_deprecated_195_auto_price	106	15	5,025	pass	0.8717	0.9485	0.9598	1.3107
a04f964bc53281e0	strogatz_lv1	267	2	985	pass	-3.8784	-3.4212	—	—
a23f4bd626b76590	feynman_test_13	800	5	1,740	pass	0.9178	0.9605	0.9900	1.0991
a32a69e7c701eb14	588_fri_c4_1000_100	667	100	19,019	fail	—	—	—	—
a5ba682572f53add	602_fri_c3_250_10	167	10	2,900	pass	0.5355	0.8109	0.7988	—
a863cd42be4d05d4	663_rabe_266	80	2	1,014	pass	0.9781	0.9933	0.9967	1.0521
a97065df07a47eb0	592_fri_c4_1000_25	667	25	5,655	pass	0.8791	0.9340	0.9578	1.2714
ab279eded3b3c52b	574_house_16H	800	16	4,035	pass	0.1181	0.1796	0.1795	0.3705
aebf5d861a8365b4	522_pm10	333	7	2,409	pass	0.5082	0.6149	0.4125	2.3076
b6adf79118a58cea	562_cpu_small	800	12	3,353	pass	0.6968	0.7747	0.9834	0.9307
b78d323566930bd7	583_fri_c1_1000_50	667	50	10,098	pass	0.8607	0.9444	0.9960	1.6211
b7f9774e2142a758	strogatz_shearflow2	267	2	1,019	pass	0.4707	0.5439	0.7450	1.8955
bc85856c012bcc1b	607_fri_c4_1000_50	667	50	10,209	pass	0.8856	0.9640	3.9394	—
c107ece39eb333a9d	584_fri_c4_500_25	333	25	5,638	pass	0.8062	0.8619	0.9417	—
c30ff9405c916718	579_fri_c0_250_5	167	5	1,692	pass	0.9475	1.0923	0.9961	0.9783
c4f8e81b70d33a13	590_fri_c0_1000_50	667	50	10,180	pass	0.8839	0.9172	1.0359	1.1762
c649a4cee87c4321	529_pollen	800	4	1,506	pass	1.0356	1.1012	0.9717	1.2459
cb5bf66ef2ee09ed	nikuradse_1	241	2	992	pass	0.9346	0.9373	0.9997	1.1909
cba8393879afd1dc	230_machine_cpu	140	6	2,338	pass	1.0026	1.0891	1.1718	—

id	name	rows	feat.	bytes	L3	CB n	CB $4n$	lin $4n$	MLP $4n$
ce1a79e2d658eb8a	strogatz_glider2	267	2	1,003	pass	0.6382	0.6625	1.0720	1.3756
ce837ccd24a3a18c	strogatz_bacres1	267	2	1,004	pass	0.7352	0.8309	0.9014	0.9826
cee6f811d0bf199d	feynman_II_11_3	800	5	1,728	pass	0.9385	0.9666	0.8950	1.0149
d255d3a2e1c8dbf4	feynman_III_12_43	800	2	1,028	pass	0.9920	0.9935	0.9966	0.9940
d52bfd2b6ab1c850	feynman_I_10_7	800	3	1,250	pass	0.9920	0.9932	0.9940	0.9959
d9a3408a840c9f48	560_bodyfat	168	14	4,364	pass	1.0247	1.0436	0.9930	1.3651
da290fae0d6c072f	feynman_test_10	800	3	1,267	pass	0.9955	0.9990	0.9990	1.0015
da78606950f100cd	1199_BNG_echoMonths	800	9	2,544	pass	0.9962	1.0086	0.9655	1.0694
dd16ac872ffb8673	strogatz_glider1	267	2	991	pass	0.1145	0.1723	0.9123	—
de2868ac837699fd	564_fried	800	10	2,861	pass	0.9477	0.9829	0.9543	1.4707
e08f0b9931f93686	225_puma8NH	800	8	2,439	pass	1.0639	1.1043	0.9951	2.8498
e18350fb51a9d1b0	strogatz_barmag2	267	2	1,027	pass	0.8897	0.9399	0.2855	—
e5a3a7b3668a1cc2	556_analcatdata_apnea2	316	3	1,238	pass	0.0339	0.0341	—	—
e68idd96cc63271b	573_cpu_act	800	21	4,954	pass	0.8080	0.9045	0.6593	1.1465
ed2efb82fce4b098	strogatz_vdp1	267	2	984	pass	0.5228	0.5812	1.2867	1.2401
edc9d41a298fd6f0	612_fri_c1_1000_5	667	5	1,705	pass	0.8224	0.8596	0.9670	1.2891
efd53e9c0f36d608	_deprecated_207_autoPrice	106	15	5,017	pass	0.9020	0.9357	1.0899	1.0398
f0a3d6881458f990	604_fri_c4_500_10	333	10	2,913	pass	0.6868	0.8360	0.8552	—
f24e9383331c43e2	503_wind	800	14	3,846	pass	0.9690	0.9962	1.0017	1.0367
f652ea96eff62654	strogatz_predprey1	267	2	989	pass	0.9964	1.1345	1.0218	1.4876
f8cf362e371e8960	strogatz_predprey2	267	2	1,030	pass	0.9876	0.9847	0.9718	1.0815
fc3812567b5fdd64	197_cpu_act	800	21	5,011	pass	0.7680	0.8922	0.6669	1.1253
fe6d7ac3fcd3129e	1201_BNG_breastTumor	800	9	2,427	pass	-0.6141	-0.3437	—	—

APPENDIX B

SUFFICIENCY BUDGET, ALL 100 LINEAGES

TABLE II: Sufficiency budget, all 100 lineages.

id	name	rows	feat.	bytes	L3	CB n	CB $4n$	lin $4n$	MLP $4n$
02a45777900441ce	215_2dplanes	800	10	—	—	—	—	—	—
047e62b52a3f4ede	695_chatfield_4	156	12	3,353	pass	1.0106	1.0287	1.0110	—
04c80944e971cb3e	first_principles_supernovae_zr	157	1	750	pass	0.9372	0.9454	1.0346	1.3271
0561d673cb73c27e	666_rmftsa_ladata	339	10	—	—	—	—	—	—
060b9fff8092db17	589_fri_c2_1000_25	667	25	5,734	pass	0.9237	0.9886	1.0258	1.5430
06b0312a585b436e	1196_BNG_pharynx	800	10	2,681	pass	0.8188	0.8768	1.0632	0.7968
0779d3fd5b3133a2	nikuradse_2	241	1	—	—	—	—	—	—
07f38946316c039b	593_fri_c1_1000_10	667	10	2,869	pass	0.9780	1.0222	0.9929	1.8097
0a8428ef25a69175	537_houses	800	8	2,636	pass	0.7243	0.7841	1.0073	1.0334
0d482c1dd7cc6592	210_cloud	72	5	1,855	pass	0.9561	0.8815	1.0056	0.9224
0db59fd2579600d0	665_sleuth_case2002	98	6	—	—	—	—	—	—
0f86a714ec5f3ac6	1191_BNG_pbc	800	18	4,218	pass	0.8069	0.8576	0.8783	—
167578f5deba4977	547_no2	333	7	2,474	pass	0.8821	0.9276	0.9850	1.0276
19f2c3c2d67defd7	feynman_II_11_27	800	4	—	—	—	—	—	—
19f4780b53b3fa11	678_visualizing_environmental	74	3	—	—	—	—	—	—
1a4456d1cc71a26a	feynman_test_11	800	7	2,185	pass	1.1162	1.3026	0.6721	—
1a803b27c2400dc4	feynman_III_10_19	800	4	1,522	pass	0.9698	0.9777	0.9732	0.9807
1cb97b1d7314d43d	229_pwLinear	133	10	2,598	pass	0.8275	1.0471	0.9727	—
205f4b4982b9ea49	608_fri_c3_1000_10	667	10	—	—	—	—	—	—
26eff843be35d831	581_fri_c3_500_25	333	25	5,655	pass	0.8660	0.9960	1.0900	1.6067
282f8ddb43b5e677	595_fri_c0_1000_10	667	10	2,913	pass	0.9645	1.0058	1.0029	1.1269
297b7688a045c583	603_fri_c0_250_50	167	50	10,183	pass	0.8257	1.1292	1.0240	2.8999
2c8d0985f3557d2b	feynman_II_11_28	800	2	—	—	—	—	—	—
30ca2887550f0d8d	201_pol	800	26	8,215	pass	0.6214	0.6436	0.7350	0.6603
3408a6a513c627a3	strogatz_barmag1	267	2	—	—	—	—	—	—
35a9a9b88f0e4d1	344_mv	800	10	—	—	—	—	—	—
3d3321e27d14f9aa	599_fri_c2_1000_5	667	5	1,713	pass	0.9820	1.0044	1.0441	1.6148
44eaabe4cb923766	613_fri_c3_250_5	167	5	1,708	pass	0.8605	0.9611	0.9427	—
46b25db9396c8209	227_cpu_small	800	12	3,338	pass	0.7033	0.8730	0.9142	0.9908
46c5a59e5ae42037	1193_BNG_lowbwt	800	9	2,468	pass	0.8393	0.8390	0.8798	0.8103
471f11cc2d38853a	feynman_test_11	800	4	1,472	pass	0.9340	0.9481	0.9730	0.9578
47c581af5de1136b	582_fri_c1_500_25	333	25	5,676	pass	0.8232	0.9820	0.8086	—
48b7a3bf81e5f3e5	605_fri_c2_250_5	167	25	5,696	pass	0.2608	0.5247	1.4268	—
4a354d98c965b01f	597_fri_c2_500_5	333	5	1,715	pass	0.9578	1.0001	1.0224	1.3200
4aa84fff8ad0cd26	feynman_III_13_18	800	4	—	—	—	—	—	—
51730b348b1bbd14	557_analcatdata_apnea1	316	3	1,246	pass	0.6556	0.6654	—	—
54b221668673164c	feynman_III_14_14	800	5	1,724	pass	1.0175	1.1226	1.0021	1.0727
552665868f8f72ed	strogatz_vdp2	267	2	—	—	—	—	—	—
585fef2a1cb6b33f	strogatz_lv2	267	2	1,021	pass	0.9239	1.0451	0.8041	—
642738a361f1c63a	feynman_I_12_1	800	2	1,034	pass	0.9936	0.9961	0.9993	0.9989
64c64f0f9983c48f	strogatz_bacres2	267	2	996	pass	0.9743	0.9826	0.9816	0.9923
654566a1f91408aa	4544_GeographicalOriginalofMusic	706	117	—	—	—	—	—	—
693890aae9dc860b	first_principles_supernovae_zg	162	1	766	pass	0.9415	0.9642	1.0277	1.1876
6cacb11611345a24	feynman_I_11_19	800	6	1,952	pass	0.9911	1.0176	1.0047	1.0089
7291b347fe0238fb	feynman_II_11_20	800	5	—	—	—	—	—	—
7423783f7e38bb9b	strogatz_shearflow1	267	2	1,023	pass	0.4860	0.5973	-4.3352	—
7a3d123dd555110d	598_fri_c0_1000_25	667	25	5,672	pass	0.8570	0.9206	1.0010	1.0168
7c8943b160464add	601_fri_c1_250_5	167	5	—	—	—	—	—	—
7e51711baa1b1fb1	feynman_I_12_11	800	5	1,720	pass	0.9780	0.9927	0.9910	1.0559
80589368fa78f21a	218_house_8L	800	8	2,410	pass	0.6661	0.7528	0.6830	1.0728

id	name	rows	feat.	bytes	L3	CB n	CB $4n$	lin $4n$	MLP $4n$
8164950b155f7f56	feynman_I_12_2	800	4	—	—	—	—	—	—
85a31ae409944a1b	feynman_II_10_9	800	3	1,264	pass	0.9803	0.9961	0.9881	1.0061
8d859cfa7b51a351	feynman_III_15_12	800	3	1,280	pass	0.7428	0.8572	0.8771	1.2306
923483635406b213	1203_BNG_pwLinear	800	10	—	—	—	—	—	—
9312d813229df035	690_visualizing_galaxy	216	4	1,470	pass	0.9770	0.9875	0.9843	1.0661
94a00ea67f49bf84	596_fri_c2_250_5	167	5	1,718	pass	0.8262	0.8934	—	—
94bd4520b8ceb3f0	feynman_test_12	800	5	1,745	pass	0.9677	0.9845	0.9937	1.0157
956d34f13e7af830	586_fri_c3_1000_25	667	25	—	—	—	—	—	—
9a10c15885022034	_deprecated_195_auto_price	106	15	5,038	pass	0.8965	0.9544	0.9901	1.3165
a04f964bc53281e0	strogatz_lv1	267	2	986	pass	-3.0531	-2.4502	—	—
a23f4bd626b76590	feynman_test_13	800	5	—	—	—	—	—	—
a32a69e7c701eb14	588_fri_c4_1000_100	667	100	19,041	fail	—	—	—	—
a5ba682572f53add	602_fri_c3_250_10	167	10	2,902	pass	0.7774	0.9773	0.9471	—
a863cd42be4d05d4	663_rabe_266	80	2	1,014	pass	0.9849	0.9996	1.0018	1.0573
ab7065df07a47eb0	592_fri_c4_1000_25	667	25	—	—	—	—	—	—
ab279eded3b3c52b	574_house_16H	800	16	4,076	pass	0.0805	0.1211	-0.7472	-0.3199
aebf5d861a8365b4	522_pm10	333	7	2,450	pass	0.6372	0.7959	0.6203	3.4124
b6adf79118a58cea	562_cpu_small	800	12	—	—	—	—	—	—
b78d323566930bd7	583_fri_c1_1000_50	667	50	10,098	pass	0.8817	0.9737	0.8733	0.9465
b7f9774e2142a758	strogatz_shearflow2	267	2	1,022	pass	0.7388	0.8339	-0.4873	1.5681
bc85856c012bce1b	607_fri_c4_1000_50	667	50	—	—	—	—	—	—
c107ee39eb333a9d	584_fri_c4_500_25	333	25	5,641	pass	0.8307	0.9024	0.9623	—
c30ff9405c916718	579_fri_c0_250_5	167	5	1,700	pass	0.9081	1.0750	0.9949	0.9830
c4f8e81b70d33a13	590_fri_c0_1000_50	667	50	—	—	—	—	—	—
c649a4cee87c4321	529_pollen	800	4	1,506	pass	1.0619	1.1198	0.9983	1.2801
cb5bf66ef2ee09ed	nikuradse_1	241	2	990	pass	0.9662	0.9677	0.9970	1.1953
cba8393879afd1dc	230_machine_cpu	140	6	—	—	—	—	—	—
ce1a79e2d658eb8a	strogatz_glider2	267	2	1,004	pass	0.5396	0.6099	1.1474	1.2017
ce837ccd24a3a18c	strogatz_bacres1	267	2	1,004	pass	0.7807	0.8568	0.8858	0.9570
cee6f811d0bf199d	feynman_II_11_3	800	5	—	—	—	—	—	—
d255d3a2e1c8dbf4	feynman_III_12_43	800	2	1,023	pass	0.9947	0.9964	0.9984	0.9967
d52bfd2b6ab1c850	feynman_I_10_7	800	3	1,248	pass	0.9978	0.9988	0.9994	1.0005
d9a3408a840c9f48	560_bodyfat	168	14	—	—	—	—	—	—
da290fae0d6c072f	feynman_test_10	800	3	1,257	pass	0.9681	0.9718	0.9737	0.9744
da78606950f100cd	1199_BNG_echoMonths	800	9	2,578	pass	0.9473	0.9673	0.9447	1.0028
dd16ac872ffb8673	strogatz_glider1	267	2	—	—	—	—	—	—
de2868ac837699fd	564_fried	800	10	2,864	pass	0.9826	1.0201	0.9548	1.5094
e08f0b9931f93686	225_puma8NH	800	8	2,440	pass	1.0168	1.0664	0.9433	2.7529
e18350fb51a9d1b0	strogatz_barmag2	267	2	—	—	—	—	—	—
e5a3a7b3668a1cc2	556_analcatdata_apnea2	316	3	1,241	pass	0.6282	0.6707	—	—
e68fdd96cc63271b	573_cpu_act	800	21	4,958	pass	0.8507	0.9689	0.5661	1.1177
ed2efb82fce4b098	strogatz_vdp1	267	2	991	pass	0.5397	0.6472	-0.1705	1.1154
edc9d41a298fd60	612_fri_c1_1000_5	667	5	—	—	—	—	—	—
efd53e9c0f36d608	_deprecated_207_autoPrice	106	15	5,024	pass	0.9035	0.9215	1.0781	1.0481
f0a3d6881458f990	604_fri_c4_500_10	333	10	2,918	pass	0.8895	1.0064	1.0421	—
f24e9383331c43e2	503_wind	800	14	3,877	pass	0.9992	0.9953	0.9918	1.0034
f652ea96eff62654	strogatz_predprey1	267	2	—	—	—	—	—	—
f8cf362e371e8960	strogatz_predprey2	267	2	1,030	pass	0.9865	0.9827	0.9658	1.0750
fc3812567b5fdd64	197_cpu_act	800	21	5,006	pass	0.8164	0.9529	0.6632	1.1726
fe6d7ac3fcd3129e	1201_BNG_breastTumor	800	9	—	—	—	—	—	—

APPENDIX C

BEYONDARENA FAMILIES IN THE DATASET BUCKET

TABLE III: All 142 BeyondArena families.

name	task	rows	cols	license	split
5g_energy_consumption	regression	92,629	22	MIT	grouped
acquire_valued_shoppers_challenge	binary	160,057	112	not cleared	temporal
airfoil_self_noise	regression	1,503	6	CC-BY-4.0	iid
allstate_claims_severity	regression	188,317	131	not cleared	iid
amazon_employee_access	binary	32,769	10	public-domain	iid
amex_non_iid	binary	1,249,605	191	not cleared	grouped
anes_voting_2026	binary	48,587	319	not cleared	temporal
aps_failure	binary	76,000	171	CC-BY-4.0	iid
asp_potassco_classification	multiclass	1,212	138	not cleared	grouped
audiology_diagnosis	multiclass	199	69	CC-BY-4.0	iid
bad_customer_detection	binary	1,723	14	public-domain	iid
bank_customer_churn	binary	10,000	11	public-domain	iid
bank_marketing	binary	45,211	14	CC-BY-4.0	iid
biogeographical_ancestry_prediction	multiclass	635	105	not cleared	iid
biomechanical_orthopaedic_prediction	multiclass	310	7	CC-BY-4.0	iid
bioresponse	binary	3,751	1777	public-domain	iid
blood_tests_drink_prediction	regression	345	6	CC-BY-4.0	iid
blood_transfusion	binary	748	5	CC-BY-4.0	iid
body_density_prediction	regression	252	14	not cleared	iid
california_house_prices_2020	regression	41,528	42	not cleared	temporal
cardiotocography	multiclass	2,126	24	CC-BY-4.0	grouped
churn	binary	5,000	20	public-domain	iid
cirrhosis_patient_survival_prediction	regression	161	18	not cleared	iid

TABLE III, continued

name	task	rows	cols	license	split
climate_model_weather_forecasting	regression	1,250,000	101	not cleared	temporal
clock_protein_toxicity	binary	171	1118	CC-BY-4.0	iid
coffee_rating_prediction	regression	2,369	13	not cleared	temporal
coil_2000	binary	9,822	86	CC-BY-4.0	iid
concrete_compressive_strength	regression	1,030	9	CC-BY-4.0	iid
consumer_complaints	multiclass	1,226,140	13	us-gov	temporal
cooking_time	regression	1,250,000	197	not cleared	temporal
covertime	multiclass	512,625	15	CC-BY-4.0	grouped
credit_approval	binary	690	16	CC-BY-4.0	iid
credit_card_clients_default	binary	30,000	24	CC-BY-4.0	iid
credit_g	binary	1,000	21	CC-BY-4.0	iid
customer_satisfaction_in_airline	binary	129,880	22	public-domain	iid
delivery_eta	regression	1,250,000	226	not cleared	temporal
dementia_prediction	multiclass	370	10	not cleared	grouped
diabetes_130_us	binary	71,518	45	CC-BY-4.0	iid
diamonds	regression	53,940	10	MIT	iid
drug_induced_autoimmunity_prediction	binary	597	178	CC-BY-4.0	iid
early_learning_predictors	regression	18,874	745	CC-BY-4.0	grouped
early_stage_diabetes_risk_prediction	binary	251	17	CC-BY-4.0	iid
ecoli_proteins	multiclass	327	7	CC-BY-4.0	iid
ecommerce_shipping	binary	10,999	11	public-domain	iid
electric_motor_temperature_prediction	regression	1,296,316	111	not cleared	grouped
emscad	binary	17,460	18	CC0-1.0	iid
eryhemato_squamous_disease	multiclass	366	35	CC-BY-4.0	iid
fiat_500	regression	1,538	8	CC0-1.0	iid
fitness_club	binary	1,500	7	public-domain	iid
food_delivery_time	regression	45,451	10	not cleared	iid
forensic_glass_identification	multiclass	214	10	CC-BY-4.0	iid
forest_fires	regression	517	13	CC-BY-4.0	iid
gallstone_disease	binary	319	39	CC-BY-4.0	iid
garments_worker_productivity	regression	1,197	16	CC-BY-4.0	temporal
ghanas_indigenous_intel	multiclass	10,928	11	not cleared	temporal
give_me_some_credit	binary	150,000	11	not cleared	iid
hazelnut_spread_contaminant_detection	binary	2,400	31	not cleared	iid
healthcare_insurance_expenses	regression	1,338	7	not cleared	iid
heart_disease_cleveland	binary	303	14	CC-BY-4.0	iid
heart_disease_hungary	binary	294	14	CC-BY-4.0	iid
heart_disease_va_long_beach	binary	200	14	CC-BY-4.0	iid
heart_failure_followup_survival	binary	299	13	CC-BY-4.0	iid
heloc	binary	10,459	24	not cleared	iid
hepatitis_c_prediction	multiclass	608	13	CC-BY-4.0	iid
hepatitis_survival_prediction	binary	155	20	CC-BY-4.0	iid
hiva_agnostic	binary	3,845	1519	public-domain	iid
home_credit_default_risk	binary	307,507	505	not cleared	iid
home_credit_default_stability	binary	1,224,927	712	not cleared	temporal
homeq_default_prediction	binary	5,708	13	not cleared	iid
homesite_quote_conversion	binary	260,753	296	not cleared	iid
horse_colic_survival	multiclass	344	21	CC-BY-4.0	iid
hotel_booking_demand	binary	81,418	32	CC-BY-4.0	temporal
houses	regression	19,675	9	not cleared	iid
hr_analytics	binary	19,158	13	CC0-1.0	iid
ieee_fraud_detection	binary	590,540	436	not cleared	temporal
immoscout_german_house_prices	regression	10,317	24	not cleared	iid
in_vehicle_coupon_recommendation	binary	12,684	25	CC-BY-4.0	iid
indian_liver_patient_dataset	binary	583	11	CC-BY-4.0	iid
iranian_churn	binary	2,850	14	CC-BY-4.0	iid
jml	binary	10,885	22	public-domain	iid
kdd_cup_09_appetency	binary	50,000	213	public-domain	iid
kick	binary	72,983	33	public-domain	temporal
kickstarter	binary	187,118	16	not cleared	temporal
labour_inspection_compliance	binary	63,634	377	CC0-1.0	iid
lending_club	binary	1,064,751	97	CC0-1.0	temporal
ljubljana_breast_cancer	binary	286	10	CC-BY-4.0	iid
ljubljana_primary_tumor	multiclass	302	18	CC-BY-4.0	iid
lung_cancer	multiclass	197	12601	not cleared	iid
lung_cancer_epithelial_genexp	binary	187	22216	public-domain	iid
maps_router_eta	regression	1,250,000	989	not cleared	temporal
marketing_campaign	binary	2,240	26	CC-BY-4.0	iid
maternal_health_risk	multiclass	1,014	7	CC-BY-4.0	iid
mercari_price_suggestion	regression	1,250,000	7	not cleared	iid
mercedes_benz_greener_manufacturing	regression	4,204	372	not cleared	temporal
miami_housing	regression	13,776	16	not cleared	iid
mic	multiclass	1,699	112	CC-BY-4.0	iid
mice_protein_trisomy_discriminant	multiclass	1,080	78	CC-BY-4.0	grouped
micro_mass	multiclass	571	1084	CC-BY-4.0	grouped
musk	binary	6,598	168	CC-BY-4.0	grouped
mutual_funds_india	regression	793	13	CC0-1.0	iid
naticusdroid_android_permissions_dataset	binary	7,491	86	CC-BY-4.0	iid
obesity_estimation	regression	498	15	CC-BY-4.0	iid
online_shoppers_purchasing_intention_dataset	binary	12,330	18	CC-BY-4.0	iid
otto_group_product_classification_challenge	multiclass	61,878	94	not cleared	iid
pancreatic_cancer_mouse_detection	binary	181	6773	not cleared	grouped
parkinsons_biomedical_voice_measurements	binary	195	25	CC-BY-4.0	grouped

TABLE III, continued

name	task	rows	cols	license	split
physiochemical_protein	regression	45,730	10	CC-BY-4.0	iid
polish_companies_bankruptcy	binary	5,790	65	CC-BY-4.0	iid
porto_seguro	binary	595,206	38	not cleared	iid
predict_students_dropout_and_academic_success	multiclass	4,424	37	CC-BY-4.0	iid
prostate_cancer_detection	binary	322	15155	not cleared	iid
pva_revenue_prediction_kddcup98	binary	144,095	478	not cleared	iid
qsar_aquatic_toxicity	regression	546	9	CC-BY-4.0	iid
qsar_biodeg	binary	1,054	42	CC-BY-4.0	iid
qsar_fish_toxicity	regression	908	7	CC-BY-4.0	iid
qsar_tid_11	regression	5,741	1025	public-domain	iid
regensburg_pediatric_appendicitis	binary	763	52	not cleared	iid
rossmann_store_sales	regression	844,392	16	not cleared	temporal
santander_customer_satisfaction	binary	71,080	308	not cleared	iid
santander_customer_transaction_prediction	binary	200,000	601	not cleared	iid
santander_transaction_value	regression	4,447	541	not cleared	iid
sat11_hand_algo_runtime	regression	2,960	171	not cleared	grouped
sberbank_housing_market_forecasting	regression	27,195	387	not cleared	temporal
sdss_17	multiclass	78,053	12	public-domain	iid
seismic_bumps	binary	2,584	16	CC-BY-4.0	iid
sepsis_prediction	binary	1,228,686	44	not cleared	grouped
sepsis_survival_minimal_clinical_records	binary	110,204	4	CC-BY-4.0	iid
sf_permit_time	regression	116,954	38	not cleared	temporal
south_africa_coronary_heart_disease	binary	462	10	not cleared	iid
splice	multiclass	3,190	61	CC-BY-4.0	iid
student_portuguese_performance	regression	649	31	CC-BY-4.0	iid
superconductivity	regression	21,263	82	CC-BY-4.0	iid
taiwanese_bankruptcy_prediction	binary	6,819	93	CC-BY-4.0	iid
telemonitoring_parkinsons_biomedical_voice_measurements	regression	502	21	CC-BY-4.0	grouped
thyroid_discordant	binary	3,711	27	CC-BY-4.0	iid
tour_travels_churn	binary	954	7	CC0-1.0	iid
video_game_fps_prediction	regression	12,288	40	CC-BY-4.0	grouped
video_transcoding_time_prediction	regression	68,784	20	CC-BY-4.0	grouped
website_phishing	multiclass	1,353	10	CC-BY-4.0	iid
wids_diabetes_mellitus	binary	127,358	182	not cleared	iid
wine_quality	regression	6,497	13	CC-BY-4.0	iid
wine_world_cost	regression	1,279	15	CC0-1.0	iid